



Fast pyrolysis of potassium impregnated poplar wood and characterization of its influence on the formation as well as properties of pyrolytic products



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HIGHLIGHTS

- Potassium influence was characterized during fast pyrolysis of lignocellulosic biomass.
- Potassium lowered thermal decomposition temperature of biomass.
- Char formation as well as its agglomeration were also promoted by potassium.
- Potassium facilitated chemical reactions, such as dehydration and demethoxylation.
- Aromatic compounds were identified from non-condensable gas fraction.

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ABSTRACT

TGA results indicated that the maximum decomposition temperature of the biomass decreased from 373.9 to 359.0 °C with increasing potassium concentration. For fast pyrolysis, char yield of potassium impregnated biomass doubled regardless of pyrolysis temperature compared to demineralized one. The presence of potassium also affected bio-oil properties. Water content increased from 14.4 to 19.7 wt% and viscosity decreased from 34 to 16.2 cSt, but the pH value of the bio-oil remained stable. Gas chromatography/mass spectroscopy (GC/MS) analysis revealed that potassium promoted thermochemical reactions, thus causing a decrease of levoglucosan and an increase of small molecules and lignin-derived phenols in bio-oil. Additionally, various forms of aromatic hydrocarbons, probably derived from lignins, were detected in non-condensed pyrolytic gas fractions.

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1. Introduction

Lignocellulosic biomass is considered a renewable and eco-friendly energy source that contributes to the reduction of carbon dioxide emissions. It can be utilized for electricity, heat, bio-fuels and chemical materials via thermochemical processes such as pyrolysis, gasification and combustion. Among those processes, fast pyrolysis has been the focus of promising methods to obtain liquid fuel, referred to as 'bio-oil' (Mohan et al., 2006). During the fast pyrolysis process, polymeric constituents of biomass are thermally degraded into numerous small molecules and converted mostly to volatile vapors that can be condensed to form bio-oil, char and non-condensable gas. The main product, bio-oil, is a complex mixture of water, solid particles and hundreds of organic

compounds that belong to acids, alcohols, ketones and multifunctional compounds (Kim et al., 2011; Milne et al., 1997). Bio-oil has several potential uses including heat and power for boilers, diesel engines and gas turbines, as well as transportation fuels and chemical resources (Czernik and Bridgwater, 2004). A pyrolysis co-product, bio-char, is also available to use for enhancement of soil quality, as a precursor of activated carbon (Azargohar and Dalai, 2006) and as a pollutant absorbent (Liu et al., 2011).

In general, the yield and physicochemical properties of pyrolytic products are closely related to future industrial applications. A large number of studies have reported that these properties are intensely affected by various factors of not only biomass feedstock but also processing conditions, i.e. heating rate, pyrolysis temperature and residence time (Demirbas, 2004; Kim et al., 2011; Onay, 2007). One important feature of biomass is the presence of inorganic metals such as K, Mg, Ca and Na, etc. The amount of inorganic metals ranges from less than 1% in lignocellulosic biomass to 15%

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in herbaceous biomass, depending on the type of biomass and its growing conditions (Agblevor and Besler, 1996). These inorganic metals, which indigenously exist in biomass, also influence the characteristics of pyrolytic products.

Previous studies have shown that inorganic metals may act as catalysts in the thermal behavior of biomass, char formation and pyrolysis product distribution (Eom et al., 2011, 2012a; Mourant et al., 2011; Nowakowski and Jones, 2008; Wang et al., 2010). These inorganic metals in biomass enhance char formation and the decomposition of biomass at lower temperatures. In addition, degradation of levoglucosan, an anhydrosugar formed from carbohydrates, is dramatically promoted (Eom et al., 2012a). Inorganic metals also have an effect on the physicochemical properties of bio-oil, especially viscosity, water content and chemical components (Mourant et al., 2011). Moreover, bio-oil and bio-char produced from fast pyrolysis of biomass are bound to contain high levels of inorganic metals. Accordingly, estimation of their contents in pyrolytic products could be important for their practical use and storage stability (Agblevor et al., 1994).

Therefore, it is essential to understand not only the catalytic effect of potassium but also its behavior during pyrolysis, in order to ensure optimal production and quality of pyrolytic products. However, there is still a lack of analysis regarding the fundamental reactions between inorganic metals and biomass and most of studies have focused on analytical pyrolysis, not on fast pyrolysis which is a practical process for production of bio-oil. In addition, there is even less information about metal distributions among pyrolytic products depending on temperature. In this study, potassium, which is abundant in biomass cell walls, was chosen as a representative inorganic metal, and potassium impregnated yellow poplar woods were prepared at various concentrations. Research focused on the investigation of the catalytic effects of potassium on pyrolysis behavior and on the characterization of essential pyrolytic products. Furthermore, distributions of inorganic metals in the pyrolytic products were also examined to study the behavior of inorganic metals during pyrolysis.

2. Methods

2.1. Feedstock

Yellow poplar wood (*Liriodendron tulipifera*: YP) was used in this experiment as raw lignocellulosic biomass, which was kindly provided by the Korea Forest Research Institute, Seoul, Republic of Korea. The size of the biomass used was approximately 0.5 mm. The wood sample was dried in an oven at 75 °C overnight prior to pyrolysis, and the final moisture contents of the samples were ~2 wt%.

2.1.1. Sample preparation

Inorganic elements in the cell wall of YP were removed by dilute hydrofluoric acid (HF, extra pure grade purchased from DUKSAN; demineralization). The demineralization process involved stirring a mixture of 3 wt% HF solution and 100 g of YP powder at 400 rpm for 1 h at room temperature. HF-treated YP was thoroughly rinsed with deionized water and then dried at 75 °C overnight (HF-YP).

To impregnate the demineralized biomass with potassium in this study, KCl ($\geq 99.0\%$, powder; Sigma-Aldrich) was used. HF-YP were immersed in KCl solution in the ratio of 10 ml/g biomass with increasing concentrations of salt at 0, 0.5, 1.0 and 2.0 wt% based on the biomass samples. Each mixture was placed on the magnetic stirrer at 200 rpm for 48 h at room temperature. After stirring, the sample was then filtered and dried in an oven at 75 °C overnight. The biomass sample treated with 0 wt% of salt

was called demineralized YP (**D-YP**) and the others were called **K-0.5**, **K-1.0** and **K-2.0**.

2.1.2. Fundamental analysis of the samples

Elemental analysis (C, H and N) was performed using a US/CHNS-932 (LECO Corp., USA). Oxygen content was calculated by difference. Holocellulose, lignin and ash contents of the samples were obtained using NREL standard procedures (Sluiter et al., 2005, 2008). The amounts of inorganic elements in raw and potassium impregnated samples were measured by inductively coupled plasma-emission spectroscopy (ICP-ES) analysis, as described in a previous study (Eom et al., 2011). Prior to fast pyrolysis, the thermal decomposition behaviors of YP and potassium impregnated samples were determined by thermogravimetric and differential thermogravimetric analysis (TG and DTG analysis) using the Q-5000 IR instrument (TA Instrument, USA). Samples (~3.5 mg) were heated under N₂ gas flow at a heating rate of 10 °C/min in the range of 40–800 °C.

2.2. Fast pyrolysis

Fast pyrolysis of raw and potassium impregnated biomasses was carried out using a fluidized bed-type pyrolyzer, which has been thoroughly described in previous studies (Eom et al., 2012b; Kim et al., 2011). Pyrolysis was operated in duplicate under temperature conditions of 450, 500 and 550 °C. Residence time was adjusted to 1.3 s by inert nitrogen gas flow. Before the operation, the pyrolyzer was purged for 30 min with nitrogen to create an oxygen-free atmosphere. A sample was fed into the pyrolyzer in a rate of 2.5 g/min. Bio-oil and bio-char, which were the target products for analysis in this study, were recovered from each collector and the yield of pyrolytic products was calculated on a wet basis. The yield of char was calculated the sum of char in the reactor and char in the char collector because it usually remained in the reactor for potassium impregnated samples.

In addition, two extracting bottles filled with acetone were connected next to the electrostatic precipitator (see Supplementary Fig. A). The non condensable gas fraction formed during pyrolysis passed through them in order to extract acetone-soluble pyrolytic products contained in non-condensable gas. These products were called '**non-condensed pyrolytic compounds**' in this study.

2.3. Characterization of pyrolytic products

2.3.1. Elemental analysis and higher heating value

Elemental analysis of bio-oil and bio-char was performed according to the same method mentioned above (Section 2.1.2) and their higher heating values (HHV) were estimated according to Sheng and Azevedo's (Sheng and Azevedo, 2005) correlation, as below:

$$\text{HHV (MJ/Kg)} = -1.3675 \times 0.03137 \times \text{C} + 0.7009 \times \text{H} + 0.0318 \times \text{O} \quad (1)$$

2.3.2. Inorganic metals

To determine the behavior of inorganic metals during fast pyrolysis, their distribution in pyrolytic products was observed by quantifying the inorganic content of bio-oil and bio-char according to the same method described above (Section 2.1.2). Since some sand, which was the fluidized bed material in the pyrolyzer, passed over into the char collector with nitrogen gas during fast pyrolysis, bio-char was mixed with sand and was difficult to separate. Accordingly, ICP-ES analysis of sand was also performed as a control for sand-mixed bio-char and then the inorganic content of bio-char was calculated considering the amount of inorganic metals derived from sand by the weight of sand in each mixture.

2.3.3. Water content, viscosity and acidity

The water content of bio-oil was determined by the standard method of ASTM E203 using a Karl-Fischer titration (DAIHAN, Korea) with Hydranal®-Composite 5 solution. The viscosity and acidity of bio-oil were measured using a capillary-type viscometer at 40 °C and a pH meter (Thermo Fischer Scientific Inc., USA), respectively.

2.3.4. Gas chromatography/mass spectroscopy (GC/MS) analysis

GC/MS analysis was performed to identify chemical compounds in bio-oil and non-condensed pyrolytic compounds dissolved in an acetone trapper. First, in order to analyze bio-oil, the samples (0.45 g) were diluted in 0.5 ml of acetone and an internal standard of 0.05 ml flouranthene was added. Quantitative determination was conducted using a gas chromatograph (Agilent 7890A) equipped with a flame ionization detector (FID) and a DB-5 capillary column (60 m × 0.25 mm × 0.25 μm). The oven was programmed to maintain 50 °C for 5 min, ramp at 3 °C/min to 280 °C and hold at that temperature for 10 min. The injector and FID detector temperatures were 250 and 300 °C, respectively, and the injector split ratio was set at 20:1. Chemical compounds in bio-oil were qualified using a mass selective detector (Agilent 5975C) with the NIST (National Institute of Standards and Technology) mass spectral library. For quantitative analysis, response factors (Rf) determined in a previous study (Eom et al., 2012b) were selectively used and Rf values of the other compounds were semi-quantitatively estimated based on their structural similarity. The concentration for compounds in bio-oil was calculated using the following equation:

Concentration for chemical compounds

$$(\text{mg/g biomass}) = Rf \times \frac{A_{\text{oil}}}{A_{\text{i.s}}} \times \frac{Q_{\text{i.s}}}{Q_{\text{oil}}} \times \frac{Y}{100} \quad (2)$$

Rf is the response factor, A_{oil} is the area of each compound, $A_{\text{i.s}}$ is the area of internal standard, $Q_{\text{i.s}}$ is the quantity of internal standard (mg), Q_{oil} is the quantity of bio-oil fraction (g) and Y is the yield of bio-oil (wt%).

3. Results and discussion

3.1. Characteristics of biomass samples

Table 1 shows the elemental and component analysis of raw and potassium impregnated biomass samples. Data indicate that the removal of inorganic metals by HF did not result in any significant changes in biomass structure or composition. Ash content of each sample increased in proportion to impregnated concentration of potassium.

Fig. 1 shows the content of inorganic elements in yellow poplar wood before and after potassium impregnation. For the raw biomass sample, more than 90 wt% of total inorganic elements consisted of potassium, magnesium and calcium, referred to as alkali

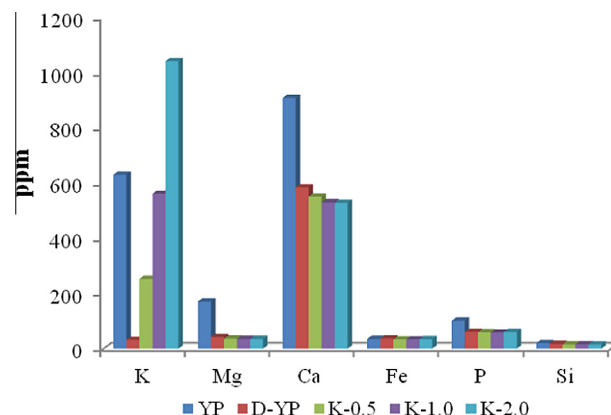


Fig. 1. Contents of inorganic elements in yellow poplar wood after potassium impregnation.

and alkali earth metals. As shown in Fig. 1 it was confirmed that inorganic elements, especially alkali and alkali earth metals were effectively removed by HF and DI water. Calcium, however, remained more than half of original content. This suggests that calcium could be strongly combined with biomass, so it was difficult to remove under the mild conditions of room temperature and a dilute acid solution. Otherwise, potassium was effectively removed and then impregnated in biomass at various concentrations.

3.2. Thermal decomposition behaviors of potassium impregnated samples

TG and DTG results comparing the effect of potassium in biomass samples are shown in Table 2. After the overall process was finished at 800 °C, 93.2 wt% of YP was volatilized and 6.8 wt% was converted to char. Potassium significantly increased the char yield from 4.4% to 9.1%, whereas volatile yield decreased from 95.6% to 90.9% for D-YP and K-2.0, respectively. The maximum decomposition temperature increased from 356.6 to 373.9 °C after the removal of inorganic metals in biomass and then it decreased to 359.0 °C as the impregnated concentration of potassium increased. These results indicated that potassium allowed thermal decomposition of biomass to occur at lower temperatures than those of the D-YP (Kowalski and Wokaun, 2007). Meanwhile, the maximum degradation rate decreased from 1.24 wt%/°C for D-YP to 1.05 wt%/°C for K-2.0. Therefore, the presence of potassium in biomass had an effect on the thermal stability of the biomass.

3.3. Yields of pyrolytic products

The yields of pyrolytic products from fast pyrolysis of raw and potassium impregnated samples are given in Table 3. The highest

Table 1
Chemical composition of raw and potassium impregnated biomasses.

	Elemental analysis (wt.%)				Holocellulose (wt.%)	Lignin (wt.%)	Ash (wt.%)
	C	H	N	O ^a			
YP ^b	45.8	5.8	0.2	48.2	74.0	30.0	0.45
D-YP ^c	47.1	6.1	0.1	46.7	72.1	29.2	0.17
K-0.5 ^d	46.4	6.3	0.1	47.2	74.1	30.0	0.22
K-1.0 ^d	47.2	6.2	0.1	46.5	74.1	28.2	0.26
K-2.0 ^d	46.9	6.3	0.1	46.7	75.0	29.1	0.37

^a By difference.

^b Yellow poplar.

^c Demineralized yellow poplar.

^d Potassium impregnated sample into HF-YP with concentrations of KCl 0.5, 1.0, 2.0 wt.%, respectively.

Table 2Pyrolysis properties of potassium impregnated biomass by TG and DTG in N₂ at heating rate of 10 °C/min.

	Yield (wt.%)		Max. degradation rate (wt.%/°C)	Temperature at max. rate (°C)
	Volatiles	Char		
YP	93.2	6.8	1.00	356.6
D-YP	95.6	4.4	1.24	373.9
K-0.5	93.7	6.3	1.10	373.1
K-1.0	93.4	6.6	1.06	368.4
K-2.0	90.9	9.1	1.05	359.0

Table 3

Yield of pyrolytic products from raw and potassium impregnated biomass under various pyrolysis temperatures.

	Bio-oil			Bio-char			Gas		
	450 °C	500 °C	550 °C	450 °C	500 °C	550 °C	450 °C	500 °C	550 °C
<i>Yield of pyrolysis products (wt.%)</i>									
YP	59.7(0.6) ^a	52.6(0.3)	46.8(0.5)	9.6(0.4)	8.4(0.1)	5.8(0.0)	30.7(1.0)	39.0(0.4)	47.5(0.5)
D-YP	59.6(1.5)	55.5(1.7)	46.7(3.6)	6.2(0.3)	4.7(0.3)	3.3(0.7)	34.2(1.7)	39.8(2.0)	50.0(2.9)
K-0.5	55.3(4.9)	53.1(3.5)	45.7(2.0)	8.1(0.5)	5.6(1.3)	5.3(0.1)	36.7(5.4)	41.3(2.2)	48.9(2.1)
K-1.0	49.8(3.1)	46.1(1.2)	44.6(4.7)	8.5(0.3)	7.0(1.4)	6.8(1.2)	41.7(3.4)	46.9(2.6)	48.6(3.4)
K-2.0	49.3(2.6)	50.0(0.2)	48.1(4.1)	12.1(0.9)	8.5(1.0)	7.3(1.7)	38.7(1.7)	41.5(1.2)	44.6(5.8)

^a Standard deviation.

yield of bio-oil, ~59.7%, was obtained at 450 °C and the bio-oil yield was critically influenced by pyrolysis temperature rather than potassium content in the biomass, as described below. The yields of bio-oil from potassium impregnated biomass under temperature conditions of 450 and 500 °C decreased due to undesirable problems occurring during fast pyrolysis processes. Under those conditions (450, 500 °C and potassium impregnated samples), char was obtained in flocculated form rather than loose particles, which is the general form of char obtained from fast pyrolysis of lignocellulosic biomass. Moreover, most char was present in the pyrolysis reactor, not in the char collector. It was experimentally ascertained that the proportion of char recovered from the reactor was approximately 57–98% for potassium impregnated samples in comparison to about 4% for YP under the same conditions. These phenomena could lead to subsequent problems such as deterioration of fluidization conditions and heat transfer rates, which are closely related to the residence time and conversion efficiency of volatiles. Thus, yields of bio-oil under those conditions

varied due to instability in the fast pyrolysis process caused by agglomeration of char in the reactor. Conversely, fast pyrolysis that was operated at 550 °C ran smoothly for all samples because of high temperature enough to lessen char formation.

Agglomeration of char induced by inorganic metals was reported in a previous study (Skrifvars et al., 1998). When inorganic metals are released, they react with accessible surfaces like fluidized bed material, usually silica, and form low-melting eutectic mixtures of inorganic metals and silica. A liquid layer on the bed material particles would tend to induce agglomeration. Additionally, char formation reactions were largely influenced by potassium in this study. Char yield increased with potassium content and even yield in the case of K-2.0 was almost double compared to that of D-YP at all temperature conditions. These data were in agreement with results of TG/DTG analysis. As temperature rose, however, char yield decreased while gas yield increased due to secondary cracking reactions promoted at high pyrolysis temperatures (Bridgwater et al., 1999; Kim et al., 2011).

Table 4

Physicochemical properties of bio-oil and bio-char.

Sample	Temperature (°C)														
	450					500					550				
	YP	D-YP	K-0.5	K-1.0	K-2.0	YP	D-YP	K-0.5	K-1.0	K-2.0	YP	D-YP	K-0.5	K-1.0	K-2.0
<i>Bio-oil</i>															
Elemental analysis(%)															
Carbon	46.9	46.5	46.3	44.9	44.7	46.2	46.7	46.4	45.4	45.0	46.1	47.0	46.3	45.3	45.0
Hydrogen	6.9	6.9	6.6	7.2	7.5	7.3	7.0	7.0	6.9	7.1	6.7	7.0	7.3	7.5	7.4
Nitrogen	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.1	0.2	0.2	0.2	0.3	0.2	0.3
Oxygen ^a	46.0	46.4	46.9	47.7	47.6	46.3	46.1	46.4	47.6	47.7	47.0	45.8	46.1	47.0	47.3
HHV(MJ/kg)	19.6	19.5	19.3	19.3	19.4	19.7	19.7	19.6	19.2	19.2	19.3	19.7	19.7	19.6	19.4
Water content (%)	14.9	14.4	13.7	15.0	16.1	15.8	14.5	14.9	16.7	18.3	17.2	14.4	15.6	17.1	19.7
Viscosity (cSt)	28.9	39.5	38.7	32.0	29.5	28.9	45.0	30.9	17.0	20.0	18.4	34.0	27.4	17.1	16.2
pH	2.9	2.5	2.6	2.5	2.2	3.0	2.6	2.7	2.6	2.4	3.1	2.6	2.7	2.6	2.3
<i>Bio-char</i>															
Elemental analysis(%)															
Carbon	74.6	77.5	79.7	77.6	78.1	75.4	75.8	75.0	79.8	81.7	80.3	79.2	81.7	79.5	81.4
Hydrogen	3.6	2.1	3.4	2.7	3.5	3.2	3.2	2.3	3.3	3.1	3.0	2.8	1.1	3.0	2.4
Nitrogen	0.5	0.7	0.5	0.4	0.4	0.5	1.2	0.3	0.6	0.4	0.5	0.9	1.3	0.8	0.5
Oxygen ^a	21.3	19.7	16.4	19.3	18.0	20.9	19.8	22.4	16.3	14.8	16.2	17.1	15.9	16.7	15.7
HHV(MJ/kg)	25.2	25.0	26.5	25.5	26.2	25.2	25.3	24.5	26.5	26.9	26.4	26.0	25.5	26.2	26.3

^a By difference.

3.4. Effect of potassium on the characteristics of pyrolytic products

3.4.1. Physicochemical properties of pyrolytic products

Table 4 illustrates physicochemical properties of bio-oil and bio-char from raw and potassium impregnated samples. As shown in Table 4, the ratios of carbon and oxygen in the bio-oil ranged from 44.7% to 47.0% and 45.8% to 47.7%, respectively. As potassium content increased from D-YP to K-2.0, the ratio of carbon in the bio-oil slightly decreased while the ratio of oxygen increased at all temperature conditions. Contrary to bio-oil, the ratio of carbon in bio-char increased while the ratio of oxygen decreased. The dehydration reaction was likely promoted by potassium during fast pyrolysis. Carbon crystallization of bio-char was also enhanced by potassium through interaction with oxygen in the bio-char to form carbon monoxide (Tsubouchi et al., 2003). In addition, an increase of the ratio of carbon in bio-char was observed as the pyrolysis temperature increased. These increases resulted from the breaking of weaker bonds within the bio-char structure at high temperatures, which led to losses of hydrogen and oxygen in bio-char (Demirbas, 2004).

The higher heating values (HHV) of bio-oil were estimated to be 19.2–19.7 MJ/kg, which are lower than those of heavy oil (45 MJ/kg) from fossil fuel. They could be improved by an upgrading process such as hydrodeoxygenation, catalytic cracking, emulsification, etc. (Bridgwater, 2012). The HHVs of bio-char, however, were estimated to be 24.5–26.9 MJ/kg, which are values similar to those of coal (25 MJ/kg). The pH values of bio-oil varied from 2.2 to 3.1. The acidity of bio-oil was due to several acids (e.g., acetic acid) that were formed during pyrolysis (Lu et al., 2009). In general, acidity of the bio-oils did not significantly change, indicating that potassium had little effect on the acidity of bio-oil.

As shown in Table 4, high potassium content in biomass resulted in high water content of bio-oil. It has been reported that dehydration reactions occur from free hydroxyl groups and carboxylic groups in carbohydrates by potassium acting as a Lewis acid (Wang et al., 2010). Jakab et al. (1993) also suggested that dehydration was promoted by the addition of alkali metals from terminal hydroxyl and aldehyde groups in lignin. Alkali metal ions in conjunction with those functional groups in the biomass were likely to be released by involving a hydroxyl group and a hydrogen atom

Table 5
Comparison of concentration of chemical compounds in bio-oil pyrolyzed at 450 °C.

No.	Source	Compound	Concentration (mg/g biomass)				
			YP	D-YP	K-0.5	K-1.0	K-2.0
1	C ^a	Acetic acid	35.78	36.71	37.05	35.03	43.55
2	C	1-Hydroxy-2-Propanone	1.90	2.42	1.79	1.42	2.34
3	C	3-Hydroxy-2-Butanone	0.20	1.74	0.31	0.23	0.14
4	C	3-Furaldehyde	0.34	0.49	0.32	0.25	0.23
5	C	Furfural	5.54	4.97	3.98	5.80	6.26
6	C	2-Cyclopentene-1,4-dione	0.60	0.21	0.60	0.62	0.28
7	C	2-Methyl-2-Cyclopenten-1-one	0.50	0.65	0.38	0.38	0.42
8	C	2(5H)-Furanone	9.94	2.60	1.08	9.52	9.08
9	C	1,2-Cyclopentanedione	1.81	2.06	3.08	1.99	4.72
10	C	5-Methyl-2(5H)-Furanone	0.71	0.51	0.76	0.65	0.73
11	C	5-Methyl-2-Furancarboxaldehyde	0.82	0.99	0.80	0.82	1.10
12	C	2H-Pyran-2-one	0.29	0.55	0.37	0.32	0.25
13	L ^b	Phenol	0.21	0.35	0.31	0.25	0.28
14	C	3-Methyl-1,2-Cyclopentanedione	1.61	0.86	0.98	1.17	1.36
15	C	4-Methyl-5H-Furan-2-one	0.91	0.39	0.72	0.76	0.57
16	L	o-Cresol	0.15	0.10	0.22	0.15	0.19
17	L	m-Cresol	0.24	0.31	0.30	0.21	0.37
18	L	Guaiacol	0.85	0.20	0.44	0.72	0.69
19	L	2-Ethyl-Phenol	0.42	0.10	0.06	0.25	0.08
20	L	4-Methyl Guaiacol	0.69	0.24	0.54	0.57	0.44
21	L	Catechol	1.66	0.84	2.25	1.75	1.88
22	L	3-Ethyl-5-methyl-Phenol	0.18	0.19	0.11	0.06	0.13
23	L	3-Methyl Catechol	0.68	0.42	0.88	0.70	0.82
24	L	4-Ethyl Guaiacol	0.50	0.23	0.21	0.27	0.40
25	L	4-Methyl Catechol	0.21	0.10	1.03	0.64	1.00
26	L	4-Vinyl Guaiacol	0.95	0.42	0.53	0.70	0.52
27	L	Syringol	0.99	0.47	0.56	0.86	0.98
28	L	Eugenol	0.34	0.49	0.20	0.35	0.27
29	L	3-Methoxy Guaiacol	0.37	0.10	0.24	0.40	0.43
30	L	4-Ethyl Catechol	0.69	0.34	0.93	0.40	0.76
31	L	Vanillin	0.47	0.18	0.33	0.34	0.29
32	L	4-Methyl Syringol	1.39	1.33	0.55	1.50	0.65
33	L	Isoeugenol (trans)	1.20	0.39	0.52	0.90	0.72
34	L	Acetoguaiacone	0.28	0.00	0.17	0.27	0.22
35	C	Levogluconan	17.68	119.38	44.78	25.62	21.21
36	L	4-Vinyl Syringol	2.35	1.49	0.40	1.03	1.47
37	L	cis-4-propenyl Syringol	0.50	0.00	0.25	0.51	0.28
38	L	Coniferyl alcohol	1.22	0.00	0.00	0.67	0.62
39	L	trans-4-Propenyl Syringol	0.37	0.00	0.00	0.28	0.16
40	L	Syringaldehyde	0.97	0.00	0.61	0.81	0.68
41	L	methoxy Eugenol	1.88	0.00	0.68	1.39	0.93
42	L	Acetosyringone	0.60	0.00	0.36	0.51	0.45
43	L	Coniferyl aldehyde	0.48	0.00	0.00	0.33	0.37
44	L	Sinapaldehyde	0.34	0.00	0.17	0.26	0.22
		Total	99.81	182.84	109.87	101.65	108.55

^a Carbohydrate derivatives.

^b Lignin derivatives.

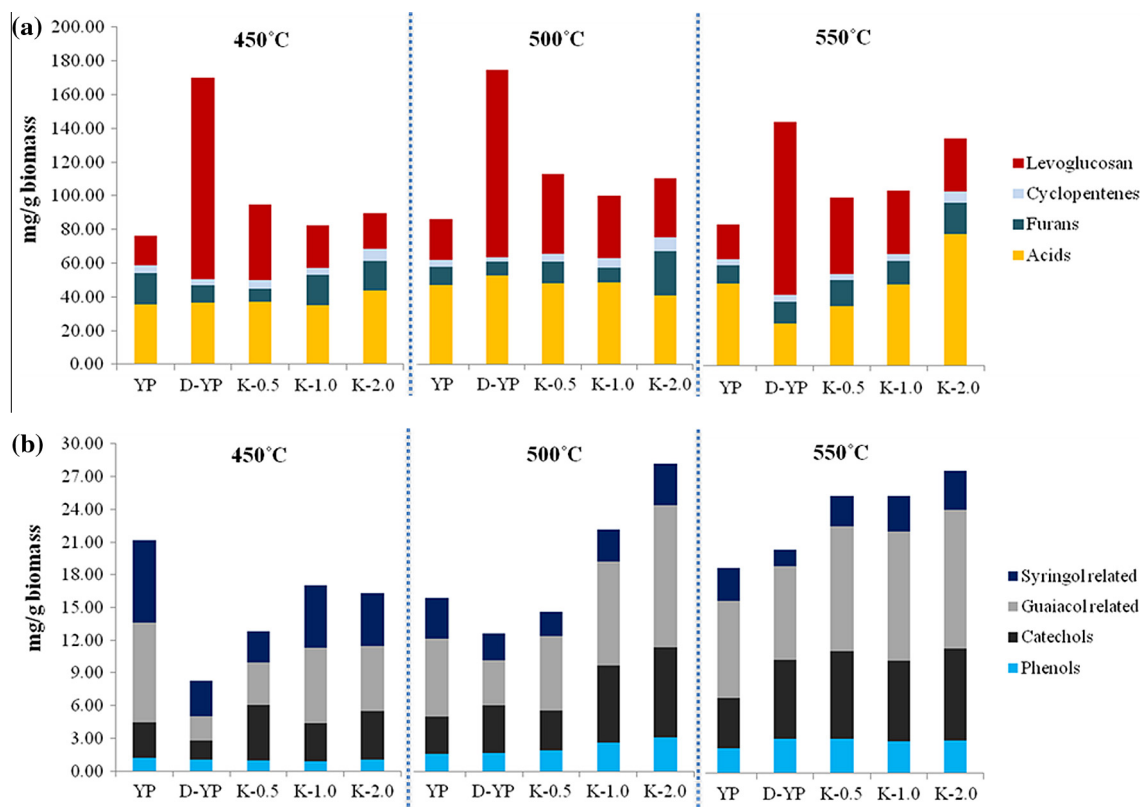


Fig. 2. Amount of different groups of chemical compounds derived from carbohydrates (a) and lignin (b).

from a near carbon to form a complex of alkali metal ion-water (H_2O-M^+ , M: alkali metal) during pyrolysis of the biomass. This complex then created a transition state and dehydration would proceed to stabilize the state (Nimlos et al., 2003). Consequently, potassium induced dehydration reactions during decomposition of the biomass.

Moreover, these dehydration reactions induced by potassium seemed to be increasingly activated as temperature increased. For instance, water content increased from 14.4 in D-YP to 16.1 wt% in K-2.0 at 450 °C, while it increased from 14.4 to 19.7 wt% at 550 °C. Different activations of potassium as a catalyst depending on temperature could be explained by the release behavior of potassium under different temperatures, as described in Section 3.5.

The viscosity of bio-oil varied from 16.2 to 45.0 cSt and this appeared in contrast to water content. Bio-oil with high viscosity was produced when inorganic metals were removed and the pyrolysis temperature was lower.

3.4.2. Chemical compounds in bio-oil

Composition of bio-oil with various potassium impregnated concentrations showed dramatic changes in the properties of bio-oil. A total of 44 compounds were identified (Table 5). The chemical compounds in the bio-oils produced from raw and potassium impregnated biomass samples were grouped by their structural characteristics e.g., levoglucosan (35), cyclopentenenes (6, 7, 9, 14), furans (4, 5, 8, 10, 11, 15) and acetic acid (1) for carbohydrate-derived compounds and syringol-related (27, 32, 36, 37, 39, 40, 42, 44), guaiacol-related (18, 20, 24, 26, 28, 29, 31, 33, 34, 38, 41, 43), catechols (21, 23, 25, 30) and phenols (13, 16, 17, 19, 22; numbers from Table 5) for lignin-derived compounds. Amounts of each group are shown in Fig. 2. The effects of potassium on bio-oil composition, especially levoglucosan, are described below.

Several studies suggested that bonds of inorganic metals and oxygen-containing functional groups in biomass could function as bridges between molecules (Chaiwat et al., 2008; Mourant et al., 2011). Thus, intermolecular cross-links would be formed and affect the thermal decomposition of biomass. Mourant et al. (2011) determined that an extended holding time by re-formation of bonds between inorganic metals and biomass components would promote secondary cracking and condensation reactions.

Furthermore, the conjugated structure could be decomposed to smaller fractions by accelerated intramolecular cleavage of its binding sites, such as the dehydration reactions mentioned in Section 3.4.1. Subsequently, ring-opened and broken structures would undergo further decomposition or recombine with released potassium leading to secondary reactions. On the other hand, a small amount of potassium in the biomass would result in fewer cross-links in the biomass structure. Cleavage of intermolecular bonds such as glycosidic linkage would likely occur and larger fractions would be released, accordingly.

Therefore, higher amounts of levoglucosan, anhydrosugar formed by cleavage of glycosidic linkages, were produced from the bio-oil with the lowest amount of potassium. Levoglucosan was, however, decomposed by interactions with potassium and then smaller amounts were present in bio-oil of K-2.0 (Fig. 2 (a)). For instance, the amount of levoglucosan decreased from 119.38 in D-YP to 21.21 mg/g in K-2.0 at 450 °C. In addition, guaiacol-related compounds dramatically increased while syringol-related compounds slightly increased with an increment of potassium (Fig. 2 (b)). These results indicated that demethoxylation from an aryl group of lignin was enhanced by potassium during pyrolysis (Jakab et al., 1993).

Temperature also influenced the chemical compounds in bio-oil. Small fractions of both carbohydrates and lignin increased at high temperature (Fig. 2). As temperature increased from 450 °C to 550 °C, guaiacol-related compounds increased from 2.25 to

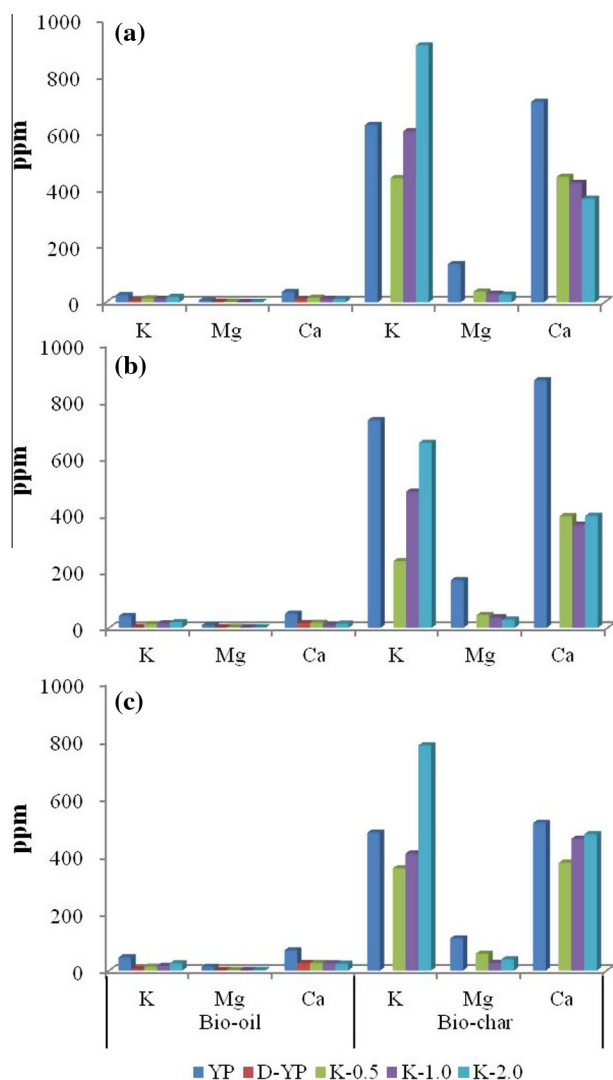


Fig. 3. Inorganic distributions in bio-oil and bio-char ((a): 450 °C, (b): 500 °C and (c): 550 °C).

Table 6

Recovery rates of inorganic metals from bio-oil and bio-char.

	450 °C	500 °C	550 °C
Recovery rate (%) ^a			
Bio-oil	3.5	4.3	5.5
Bio-char	93.9	89.6	87.7
Total	97.4	93.9	93.2

^a The ratio of inorganic concentrations between pyrolytic products and biomass.

8.59 mg/g in D-YP and from 5.90 to 12.74 mg/g in K-2.0. On the contrary, syringol-related compounds decreased from 3.29 to 1.56 mg/g in D-YP and from 4.89 to 3.58 mg/g in K-2.0. The results indicated that demethoxylation could also be facilitated by high temperature.

3.4.3. Analysis of non-condensed pyrolytic compounds

Nine non-condensed pyrolytic compounds were identified from the gas fraction occurring during fast pyrolysis and their GC/MS chromatograms and structural formulas are shown in Supplementary Fig. B. It is well known that four compounds (a, c, d, f) are derived from carbohydrates during pyrolysis (Faix et al., 1991). The other compounds (b, e, g, h, i) are aromatic hydrocarbons,

commonly called BTEXS (benzene, toluene, ethylbenzene, xylenes and styrene) and are utilized as valuable commodity chemicals (Vispute et al., 2010). Due to their aromatic structure, those compounds are assumed to be easily derived from lignin via demethoxylation and side chain cleavage (Faix et al., 1990; Nowakowski and Jones, 2008). However, those compounds were rarely reported as chemical components of bio-oil produced from lignocellulosic biomass. From this research, it is clear that BTEXS are formed during fast pyrolysis of lignin, although their concentrations were low and were not condensed to liquid form of bio-oil under the current condensation system (cooler and electrostatic precipitator).

The reason for the inability to condense BTEXS to bio-oil could be narrowly explained with their structural features (related to the electronegativity value of compounds). All chemical compounds identified from bio-oil have functional groups such as hydroxyl, methoxyl or carbonyl groups in their own structures (Table 5), whereas BTEXS do not. In the fast pyrolysis system, volatiles formed during pyrolysis immediately pass through the cooler and electrostatic precipitator. Most of the volatiles turned into bio-oil in this process, while BTEXS, unable to become bio-oil, passed through them and dissolved in acetone. Quantitative analysis of those compounds will be performed in future studies.

3.5. Distribution of inorganic metals in pyrolytic products

The contents of inorganic metals in bio-char and bio-oil were analyzed to investigate inorganic metal distributions and their behavior during pyrolysis (Fig. 3). Since most inorganic metals in the biomass were composed of K, Ca and Mg, their contents were estimated in this study. The amount of bio-char in D-YP samples was too little to conduct ICP-ES analysis. Potassium content of pyrolytic products gradually increased with impregnated concentration and there was a distinct difference of residual amounts of bio-oil and bio-char regardless of the kind of inorganic metals (Fig. 3).

Table 6 provides data on the recovery rate of total inorganic metals. It was revealed that approximately 88–94% of inorganic metals remained in the bio-char, and these results agreed with prior observations (Aglebor and Besler, 1996; Jendoubi et al., 2011). As temperature increased, however, total recovery rate gradually declined and the rate of char decreased while the rate of oil slightly increased, indicating that the release behavior of inorganic metals is related to pyrolysis temperature. It is still not clear, but inorganic metals in biomass probably exist as a form of inorganic salts or in an ionized state combined with oxygen-containing functional groups (Jensen et al., 2000). As biomass decomposes, inorganic metals are partially released from the biomass and some of them would be present in the char while others would evaporate into a gas after pyrolysis. In addition, inorganic metals in char possibly form intercalation compounds with the surface layer of char (Kapteijn et al., 1983; Sharma et al., 2004). Thus, the high temperature could enhance further decomposition of biomass as well as cleavage of weaker bonds between inorganic metals and the char surface. Consequently, this would lead to a greater release of inorganic metals, so the amount remaining in char probably decreased. Moreover, some of the inorganic metals were perhaps caught in bio-oil and then its recovery rate increased little by little at high temperature.

4. Conclusion

Fast pyrolysis of yellow poplar under various potassium concentrations and temperatures were performed to investigate catalytic effects of potassium and its behavior during pyrolysis. TGA revealed that overall decomposition temperature of biomass increased due to potassium. Properties and chemical compounds

of bio-oil as well as char formation were also largely influenced. Water content and small molecules in bio-oil increased due to dehydration, demethoxylation and secondary reactions promoted by potassium. Aromatic hydrocarbons were identified from non-condensed pyrolytic compounds trapped in acetone. Most of inorganic metals remained in bio-char and their release behavior during pyrolysis depended on temperature.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.biortech.2013.09.132>.

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